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|---|--|---|--|---|--|---|--|---|--|---|--|
| 1 |  | 2 |  | 3 |  | 4 |  | 5 |  | 6 |  |
| A |  | A |  |   |  |   |  |   |  |   |  |
| B |  | B |  |   |  |   |  |   |  |   |  |
| C |  | C |  |   |  |   |  |   |  |   |  |
| D |  | D |  |   |  |   |  |   |  |   |  |

SFM2200-010-00041DY

|     |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
|-----|---------------------------|-----------------------------------|---------------------------|----------------------------------------------------------|------|---|------|---|----|-----|---------------------|---|-------------|---------|-----------------------------------|--|
| 40  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 39  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 38  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 37  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 36  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 35  | ETB(1~3)                  | Grounding terminal box            |                           | 2                                                        | 4    | 4 | 2    | 4 | 4  | 4   | 4                   | 2 |             |         |                                   |  |
| 34  | JB(1~2)                   | Terminal box of current collector |                           | 2                                                        | 2    | 2 | 2    | 2 | 2  | 2   | 2                   | 2 |             |         |                                   |  |
| 33  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 32  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 31  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 30  | U, V, W, U, V, W, U, V, W | IGBT module for VVVF              | OCWAKC3-18U(1700,3000)    |                                                          | 6    | 6 |      | 6 | 6  | 6   | 6                   |   |             |         | Mitsubishi Electric               |  |
| 29  | OV1                       | Chopper unit of brake             | OCWAKC3-18U(1700,3000)    |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 28  | DB(1~2)                   | Diode                             | DB60W0-14S(1700,1800A)    |                                                          | 2    | 2 |      | 2 | 2  | 2   | 2                   |   |             |         | Mitsubishi Electric               |  |
| 27  | FC                        | Filter capacitor                  | 1050V, 30000μF            |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 26  | DOCTS                     | Current sensor                    | CT-40ES2P00ESGB(10MA/10V) |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 25  | DOCT1                     | Current sensor                    | CT-40ES2P00ESGB(10MA/10V) |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 24  | CTV1                      | Current sensor                    | CT-40ES2P00ESGB(10MA/10V) |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 23  | CTU1                      | Current sensor                    | CT-40ES2P00ESGB(10MA/10V) |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 22  | OVCT1                     | Current sensor                    | CT-40ES2P00ESGB(10MA/10V) |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 21  | DOCT(1~2)                 | Voltage sensor                    | EXB556(1500V/10V)         |                                                          | 2    | 2 |      | 2 | 2  | 2   | 2                   |   |             |         | Mitsubishi Electric               |  |
| 20  | LB                        | Line contactor                    | 800V, 80A                 |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 19  | CHB                       | Charge contactor                  | 750V, 60A                 |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 18  | DCHR                      | Discharge resistance              | 0.02Ω×600.0/12A×600.0/12P |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 17  | CHRe                      | Charge resistance                 | 0.06Ω×200A×0/10A×0/10S    |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 16  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 15  |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 14  | EB(1~4)                   | Grounding device                  |                           |                                                          | 4    | 4 |      | 4 | 4  | 4   | 4                   |   |             |         | Mitsubishi Electric               |  |
| 13  | ED(1~2)                   | Axle end grounding device         |                           | 2                                                        |      |   | 2    |   |    |     |                     | 2 |             |         | Mitsubishi Electric               |  |
| 12  | M(1~4)                    | Traction motor                    | 190kw, 595V, 240A         |                                                          | 4    | 4 |      | 4 | 4  | 4   | 4                   |   |             |         | Mitsubishi Electric               |  |
| 11  | VVVF                      | VVVF main inverter                |                           |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 10  | BR                        | Brake resistance                  | DC800V, 0.5Ω, 100kW       |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 9   | FL                        | Filter reactor                    | DC800V, 720A              |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 8   | HB                        | High-speed circuit breaker        | DC1000V, 1000A            |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 7   | MF                        | Main fuse                         | 800V, 600A×3              |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 6   |                           |                                   |                           |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 5   | MS                        | Main isolation switch             | DC800V, 1000A             |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 4   | BHB                       | High-speed circuit breaker of bus | DC1000V, 1500A            |                                                          |      |   | 1    |   |    |     | 1                   |   |             |         | Eisubishi Electric (only for RLK) |  |
| 3   | BF                        | Bus fuse                          | 800VDC, 600A×3            |                                                          | 1    | 1 |      | 1 | 1  | 1   | 1                   |   |             |         | Mitsubishi Electric               |  |
| 2   | BLB                       | Bus contactor box                 | DC1500V, 1100A            |                                                          |      |   | 1    |   |    |     | 1                   |   |             |         | Eisubishi Electric (only for RLK) |  |
| 1   | CS                        | Current collector                 |                           | 4                                                        | 4    | 4 | 4    | 4 | 4  | 4   | 4                   | 4 | 4           | 4       | Mitsubishi Electric               |  |
| No. | Code                      | Designation                       | Type and size             | Quantity                                                 |      |   |      |   |    |     |                     |   |             | Remarks |                                   |  |
|     |                           |                                   |                           | CSR Qingdao Sifang Co., Ltd.                             |      |   |      |   |    |     |                     |   |             |         |                                   |  |
|     |                           |                                   |                           | Detailed Diagram of High-Voltage, Traction and Grounding |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 版本  | 处数                        | 更改文件号                             | 签 名                       | 年 月 日                                                    |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 设 计 |                           |                                   | 工 艺                       |                                                          | 阶段标记 |   | 质 量  |   | 比例 |     | 编 码                 |   | 34130100042 |         |                                   |  |
| 校 核 |                           |                                   |                           |                                                          | S    |   |      | 0 |    | 1:1 |                     |   |             |         |                                   |  |
| 审 核 |                           |                                   |                           |                                                          | 共 1张 |   | 第 1张 |   |    |     | SFM2200-010-00041DY |   | 版本          |         |                                   |  |
| 主 管 |                           |                                   |                           | 标准                                                       |      |   |      |   |    |     |                     |   |             |         |                                   |  |
| 总 体 |                           |                                   | 批准                        |                                                          |      |   |      |   |    |     |                     |   |             |         |                                   |  |

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SFM2200-010-00041DY

|     |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
|-----|-----------|-----------------------------------|--------------------------|------------------------------|------|---|------|---|-----|---------------------|---|---|----------------------------------------------------------|--------------------------------|--|
| 40  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 39  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 38  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 37  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 36  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 35  | ETB(1~3)  | Grounding terminal box            |                          | 2                            | 4    | 4 | 2    | 4 | 4   | 4                   | 4 | 2 |                                                          |                                |  |
| 34  | JB(1~2)   | Terminal box of current collector |                          | 2                            | 2    | 2 | 2    | 2 | 2   | 2                   | 2 | 2 |                                                          |                                |  |
| 33  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 32  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 31  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 30  | Ⅱ、Ⅲ、Ⅳ、Ⅴ、Ⅵ | IGBT module for VVVF              | QZ40RKC3-16U(1700V,200A) |                              | 6    | 6 |      | 6 | 6   | 6                   | 6 |   |                                                          | Mitsubishi Electric            |  |
| 29  | OV1       | Chopper unit of brake             | QZ40RKC3-16U(1700V,200A) |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 28  | DB(1~2)   | Diode                             | 0R6X0B-16S(1700V,1600A)  |                              | 2    | 2 |      | 2 | 2   | 2                   | 2 |   |                                                          | Mitsubishi Electric            |  |
| 27  | FC        | Filter capacitor                  | 1050V,30000μF            |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 26  | DOCTS     | Current sensor                    | FE-40E2P010E30(10mA/10V) |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 25  | DOCT1     | Current sensor                    | FE-40E2P010E30(10mA/10V) |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 24  | CTV1      | Current sensor                    | FE-40E2P010E30(10mA/10V) |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 23  | CTU1      | Current sensor                    | FE-40E2P010E30(10mA/10V) |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 22  | OVCT1     | Current sensor                    | FE-40E2P010E30(10mA/10V) |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 21  | DOPT(1~2) | Voltage sensor                    | EX9556(1500V/10V)        |                              | 2    | 2 |      | 2 | 2   | 2                   | 2 |   |                                                          | Mitsubishi Electric            |  |
| 20  | LB        | Line contactor                    | 800V,80A                 |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 19  | CHB       | Charge contactor                  | 750V,60A                 |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 18  | DCBR      | Discharge resistance              | 30V(20±50Ω)/1(20±50Ω)/CP |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 17  | CHRe      | Charge resistance                 | 30V(±30±4Ω)/30m±0.1US    |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 16  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 15  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 14  | EB(1~4)   | Grounding device                  |                          |                              | 4    | 4 |      | 4 | 4   | 4                   | 4 |   |                                                          | Mitsubishi Electric            |  |
| 13  | ED(1~2)   | Axle end grounding device         |                          | 2                            |      |   | 2    |   |     |                     |   | 2 |                                                          | Mitsubishi Electric            |  |
| 12  | M(1~4)    | Traction motor                    | 190kw,595V,240A          |                              | 4    | 4 |      | 4 | 4   | 4                   | 4 |   |                                                          | Mitsubishi Electric            |  |
| 11  | VVVF      | VVVF main inverter                |                          |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 10  | BR        | Brake resistance                  | DC800V,0.5Ω,100kW        |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 9   | FL        | Filter reactor                    | DC800V,720A              |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 8   | HB        | High-speed circuit breaker        | DC1000V,1000A            |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 7   | MF        | Main fuse                         | 800V,600AX3              |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 6   |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 5   | MS        | Main isolation switch             | DC800V,1000A             |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 4   | BHB       | High-speed circuit breaker of bus | DC1000V,1500A            |                              |      |   | 1    |   |     |                     | 1 |   |                                                          | Kinohi Electric (only for RLK) |  |
| 3   | BF        | Bus fuse                          | 800VDC,600AX3            |                              | 1    | 1 |      | 1 | 1   | 1                   | 1 |   |                                                          | Mitsubishi Electric            |  |
| 2   | BLB       | Bus contactor box                 | DC1500V,1100A            |                              |      |   | 1    |   |     |                     | 1 |   |                                                          | Kinohi Electric (only for RLK) |  |
| 1   | CS        | Current collector                 |                          | 4                            | 4    | 4 | 4    | 4 | 4   | 4                   | 4 | 4 | 4                                                        | Mitsubishi Electric            |  |
| No. | Code      | Designation                       | Type and size            | Quantity                     |      |   |      |   |     |                     |   |   | Remarks                                                  |                                |  |
|     |           |                                   |                          | CSR Qingdao Sifang Co., Ltd. |      |   |      |   |     |                     |   |   | Detailed Diagram of High-Voltage, Traction and Grounding |                                |  |
|     |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 版本  | 处数        | 更改文件号                             | 签名                       | 年月日                          | 阶段标记 |   |      |   |     |                     |   |   |                                                          |                                |  |
| 设计  |           |                                   | 工艺                       |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 校核  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 审核  |           |                                   |                          |                              |      |   |      |   |     |                     |   |   |                                                          |                                |  |
| 主管  |           |                                   | 标准化                      |                              | S    |   |      | 0 | 1:1 | SFM2200-010-00041DY |   |   |                                                          | 版本                             |  |
| 总体  |           |                                   | 批准                       |                              | 共 1张 |   | 第 1张 |   |     |                     |   |   |                                                          |                                |  |